

CostVision

What a part should cost to make — every number traced to a rate

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Cost Engineering & Digital Innovation

19 Processes

CAD to Cost, on Rules

AI Switched Off at JLR

DFM / DFA Advice

20 Regions

Assembly Roll-Up

Learning Curve

Supplier Quote Comparison

Windows Laptop, No Internet

THE CHALLENGE

Why CostVision? Costing by Hand Is Slow

Most should-costs are built by hand, one spreadsheet at a time. That is slow, and hard to defend.

Slow

A should-cost built by hand can take weeks. By the time it lands, the sourcing decision is often made.

Quotes Taken on Trust

Without our own number, a quote is hard to challenge. The supplier's margin stays hidden.

No Common Method

Each engineer builds the cost a different way. Assumptions are not written down, so numbers cannot be compared.

Country Rates Are Hard

Labour, machine, energy and material rates differ by country. Adjusting them by hand is easy to get wrong.

Problems Found Late

Cost problems show up near launch, not at concept. By then a change means new tooling and lost margin.

No Floor Price

Buyers go into a negotiation without a price they can defend line by line.

~80%

Industry rule of thumb, not a CostVision result: most of a part's cost is fixed at the design stage, yet most cost work happens after design freeze. We want a defensible number earlier.

THE SOLUTION

What is CostVision?

A should-cost engine. Every number is arithmetic you can trace to a rate. It needs no AI.

CAD to Cost

Upload STEP, IGES or STL. The tool measures the part, and asks you where the geometry cannot decide.

Form Entry

Type the inputs for any of the 19 processes. Same engine, same eight cost buckets.

19 Manufacturing Processes

Machining to wiring harness, plus an assembly roll-up and an automotive software cost model.

Rate Library

Dated 16 June 2026: 328 materials, 178 machines, 42 labour grades. Editable and versioned.

20 Regions

Each with its own labour, energy and machine rates, shown in its own currency.

Checks Its Own Work

A self-audit and a P10–P90 uncertainty band come with every result.

Supplier Quote Comparison

Put a quote next to the should-cost, bucket by bucket, and see where the gap is.

DFM / DFA Advice

Rule-based advice on how to make the part cheaper. It never changes the price.

Exports

6-sheet Excel, PDF report and negotiation pack. Each matches the screen to the penny.

OPTIONAL AI MODE · SWITCHED OFF AT JLR

AI Mode — Built In, Switched Off, Never Sets a Price

The AI code is in the product but turned off. If AI is ever approved, a setting turns it on — no rebuild.

How It Works

1. What it would do (if switched on)

Read a plain-English description, a drawing or a PCB photo, and suggest the inputs: material, process route, features.

2. What it never does

Set a price. Every pound is calculated by the same rules engine, from the rate library.

3. How it is kept in check

Its suggestions are checked against the measured geometry. Every AI route is rate-limited per user.

4. What JLR sees today

Screens that need AI are hidden. If you reach one, the tool says "AI is switched off in this installation".

● **CostVision — JLR build, AI switched off**

You: Aluminium bracket, 6082-T6, 200 × 100 × 50 mm, three holes, 5,000 a year, UK

CostVision: AI is switched off in this installation.
Costing, CAD geometry and the learning loop work without it.

- Upload the STEP file, or
- Fill in the machining form

You: Uploads bracket.step

CostVision: Measured: volume, weight, size, walls, holes.
One question: which process route — machining or casting?
Then: cost from the rate library, in eight buckets.

At JLR

No AI, no key

Same engine either way

Works offline

On by a setting

CAD TO COST

CAD to Cost: Measured by the Tool, Checked by You

The geometry engine measures the part. Rules turn that into cost inputs. You answer what it cannot decide.

<0.01%

Engine vs hand calc,
reference part

8

Cost buckets, each traced
to a rate

6

Real parts pinned in
regression

Supported formats

.STEP .STP .IGES .IGS .STL + .DXF flat pattern for sheet metal

Measured by the OpenCASCADE geometry kernel. The DXF gives the true blank instead of the bounding box.

What the Tool Measures

- Volume and weight — weight is volume × the density of the material you pick
- Size: bounding box, wall thickness and how much of the box the part fills
- Holes, bosses, threads and other features (from STEP and IGES)
- Face types and areas, for machining and coating
- A bottom-up CNC cycle estimate built from the feature table
- Walls over ~6 mm: offered casting, forging or machining — never sheet or moulding
- 13 of the 19 processes can be costed straight from CAD this way

Where the Tool Asks You — and What Comes Out

Process route

Asked if more than one route fits

Material family

Asked: steel, aluminium, plastic ...

Hole count on an STL

Asked: an STL has no feature table

Cycle time, machined STL

Costing blocked until you type one

Service questions

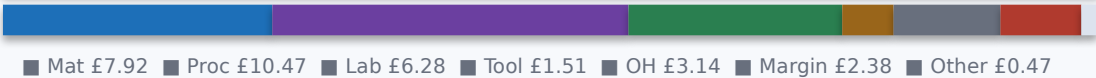
Pressure-tight? Safety-critical?

Sheet-metal blank

Measured from a DXF flat pattern

Real part: steering knuckle, forged steel, UK rates

£ 32.16



Pinned in the real-parts baseline. It records what the tool says — it has not been compared with a price JLR paid.

3D CAD VIEWER

3D CAD Viewer — Inspect, Measure and Check

Built into CostVision. STEP, IGES and STL in the browser — no separate CAD licence needed.

Views & Navigation

Iso, front, top and right views, fit to screen, a view-cube, and a zoom that holds up on large parts. Full-window mode.

Display & DFM Checks

Shaded or wireframe, bounding-box sizes, colour by face type, a colour per body with a legend, a draft and undercut map, and a wall-thickness map.

Structure & Assembly

Model tree of bodies, features and faces. Section planes on X, Y and Z. Exploded view. Move or rotate one body while the rest stays put.

Measure & Inspect

Distance, radius, diameter, angle and point position, read from the exact B-rep. Export to CSV, or snapshot into the report.

Fits the Workspace

A dockable model tree, tool groups you can drag into your own order, and a toolbar that slides away. Works beside the app's sidebar.

One viewer, many jobs

STEP / IGES / STL
Runs in the browser
Exact B-rep measurements
No extra CAD licence

The same geometry
feeds the cost engine →

CAD TO COST · MEASURED, NOT TYPED

What the Tool Measures — and What It Feeds

Geometry is measured, not typed. Every £ is arithmetic you can trace to the rate library.

Measured from the geometry (no typing)	→ Feeds this cost driver
Volume — exact, from the geometry kernel (OpenCASCADE)	Material mass · stock size
Weight = volume × density of the chosen material	Raw-material £
Bounding box / envelope	Machine sizing · stock · setups
Body / component count	Assembly scope · BOM line count
Hole & boss table — Ø, depth, count (STEP / IGES)	Secondary machining: drill / bore / tap
Wall thickness	Which routes are offered · moulding and casting feasibility
Face types and areas (planar / cylindrical)	Machining area · paint / coat area
CNC cycle-time and setup estimate	Process-time baseline for machining
DXF flat pattern (sheet metal, optional)	True blank size, instead of the bounding box

Worked example — Spur Gear (m3 · z38)

Case-hardening steel, ISO class 8 — measured:
Volume 266.0 cm³ · Steel weight 2.09 kg
Envelope 120 × 120 × 30 mm · 1 body

→ Should-cost, UK rates (pinned in the baseline):
Material £8.14 · Process £8.04 · Labour £3.61
Tooling £0.96 · Overhead £2.49 · Margin £1.89
+ packaging and logistics £0.36
= £25.50 / part

The golden rule

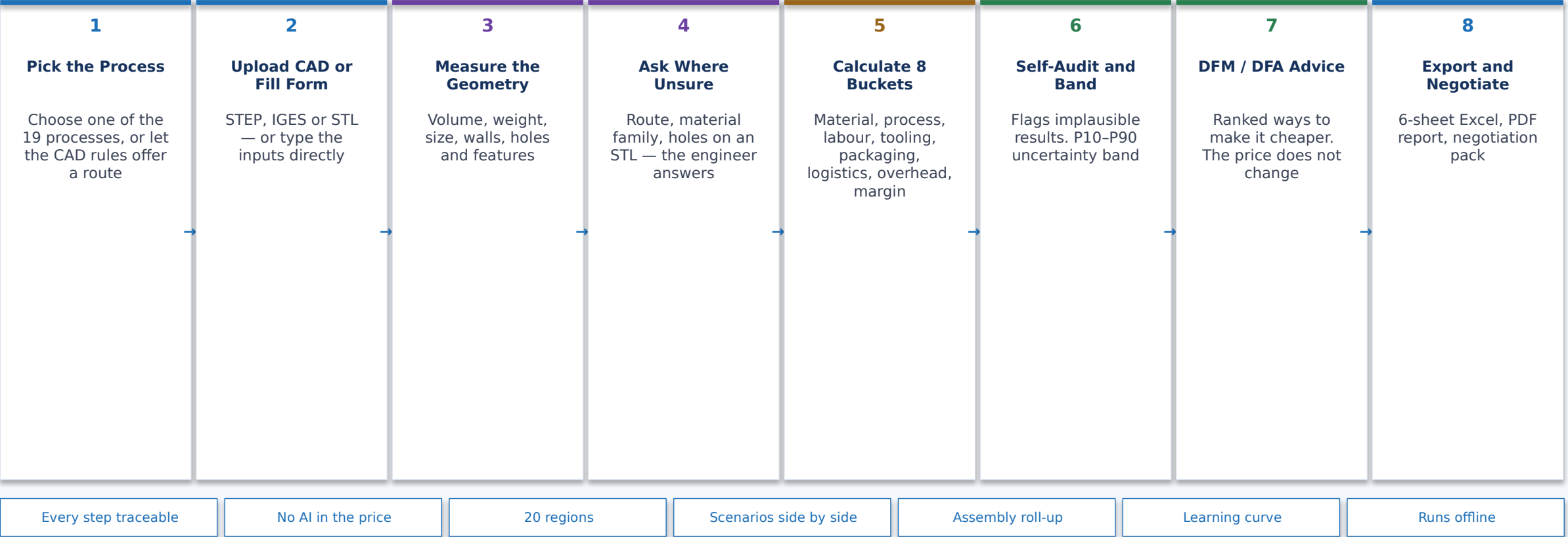
No AI in the price. At JLR the whole chain runs on measurement and rules. Even where AI is switched on, it only reads and classifies. Every number above is arithmetic, bounded by the measured geometry and traceable to the rate library.

WORKFLOW

From Part to Cost — Eight Steps

Two ways in at JLR: CAD upload or the form. The same engine either way.

Way in A: CAD upload (STEP / IGES / STL) | Way in B: Form entry | Way in C: Describe the part in words — AI mode, switched off at JLR



FULL COVERAGE

19 Manufacturing Processes — One Engine

Plus an assembly roll-up and an automotive software cost model. The same eight cost buckets for all.

01 CNC Machining Milling, turning, drilling	02 Casting HPDC, gravity, investment, sand	03 Cast + Machine Casting plus finish machining	04 Forging Hot die forging	05 Sheet Metal Pressed: progressive / transfer	06 Sheet Metal Fab Laser · punch · brake · weld	07 Gear Cutting Hob · shape · skive · grind
08 Injection Moulding Thermoplastics, multi-cavity	09 Blow Moulding Hollow parts: tanks, ducts	10 Extrusion Profile, pipe, tube	11 Thermoforming Vacuum, pressure, twin-sheet	12 Rotational Moulding Large hollow shapes	13 Rubber Moulding Compression, transfer, LSR	14 Composites Hand lay-up, RTM, autoclave
15 Painting / Coating E-coat, powder, wet, anodise	16 BIW Assembly Spot weld, MIG, hem, bond	17 PCB Fabrication Bought-in price, fabricator tables	18 PCBA SMT, through-hole, reflow	19 Wiring Harness Cut, strip, crimp, test	20 Assembly Roll-Up Multi-part BOM, one total	21 Automotive Software 49-module cost model

ENGINEERING DEPTH

What Sits Behind Every Number

Twelve layers, all of them arithmetic you can follow. Nothing is estimated by feel.

Materials 328 materials in the rate library dated 16 June 2026: metals, plastics, rubber, composites, laminates.	Machines & Labour 178 machines and 42 labour grades, each rate editable per project and per region.	Process Routing The order of operations, and for each one: which machine, how long, who runs it.	Tooling & NRE Dies, moulds and fixtures spread over annual volume × programme life.
Yield & Scrap What actually comes out good first time, driven by the features that make a part hard to build.	Regional Rates 20 regions, each with its own labour, energy and machine rates.	Benchmarks & Sensitivity Checked against typical cost ranges, and ranked by which inputs move the number most.	Learning Curve What the part costs as volume builds — the figure you need for a long-term agreement.
Quote Comparison Put the supplier's quote next to ours, bucket by bucket, in their currency or yours.	Uncertainty Band A P10–P90 range on every result, from a Monte Carlo run over every cost driver.	Gear Cutting Hobbing, shaping, skiving and grinding times worked out from the gear itself.	Landed Cost Duty, carbon levy and shipping terms — the cost at our door, not at the supplier's gate.

ADVANCED CAPABILITIES

Beyond a Cost Number

Five tools that turn the estimate into something you can negotiate with.

Learning Curve

Cost falls as volume builds. Set the curve, read the cost at any year.

- Talk about year-three cost, not first-batch cost
- Built for long-term agreements and make-vs-buy

Assembly Roll-Up

Cost a whole module, not just one part.

- Mix any of the 19 processes in one build
- Every component keeps its own cost model
- Body + paint + harness + PCBA = one total

Quote Comparison

Log the supplier's quote. Any currency, converted for you.

- Compared with the should-cost, bucket by bucket
- The gap stated plainly: "Quote is X% above should-cost"

What Moves the Number

Each driver flexed $\pm 10\%$ (by default), ranked by impact.

- Shows the few inputs worth checking before you negotiate

Scenarios Side by Side

Baseline, target and stretch on one screen.

- See exactly where the money moved
- Saved to your account to come back to

ADVICE LAYER

DFM / DFA — How to Make the Part Cheaper

Runs after the cost is settled. It reads the result and advises — it never changes the price.

Manufacturability Score (DFM)	Assembly Score (DFA)	Actions	Score drops with each issue found · each action carries its own saving estimate
1 - 10	1 - 10	Ranked	
What It Finds — Making the Part Too much material going to scrap. Machines running well below what they should. Tooling, overhead or supplier margin out of line. Each issue graded critical, major or minor.		What It Finds — Building It Assembly done by hand that could be automated. Too many setups — each one adds cost and variation. Too many different fasteners, so tools keep changing.	
What to Do About It A ranked list of actions, each with its own saving estimate: Automate the manual operations Move to a near-net shape Raise machine uptime Plus quick wins: more volume, open-book overhead, re-bid.		Checks on the 3D Model A second layer reads the CAD model itself: casting, forging, moulding, machining, sheet metal. It points at the faces that cause the problem. Every rule names its source. A finding is priced only when rates and volume are known.	

GLOBAL COVERAGE

20 Manufacturing Regions — Each in Its Own Currency

Pick a region and every rate resets together — labour, machines, energy, material and currency.

12 of the 20 regions — skilled labour, 2026 Q2

Region	Currency	Skilled labour (£/hr equiv.)	Where
UK	GBP	£26.00	Base region
Germany	EUR	£40.50	Western Europe
Poland	PLN	£12.00	Central Europe
Romania	RON	£7.50	Eastern Europe
China	CNY	£7.90	East Asia
India	INR	£5.10	South Asia
Mexico	MXN	£7.50	Near-shore for the USA
USA	USD	£34.00	North America
Thailand	THB	£5.80	South-east Asia
Vietnam	VND	£3.80	South-east Asia
Brazil	BRL	£8.50	South America
S. Korea	KRW	£22.00	East Asia

Currency Follows the Region

Pick a region and the display currency switches to its own. Choose a currency by hand and it stays put.

Every Rate Resets

Eight labour rates, electricity, gas, machine rates and material prices — all set for the region.

Editable Rate Library

Version 2.1.0, dated 16 June 2026. Override any rate. JLR's own rate card converts in with a macro-free workbook.

All 20 Regions at Once

One table shows the same part costed in every region — a make-vs-buy view built on rates.

WHAT RUNS TODAY · WHAT WAITS

Runs on Rules Today — AI Mode Waits for Approval

Left: what runs at JLR now, with AI switched off. Right: the optional AI features — built, but turned off.

Runs Today — No AI

Rules and arithmetic. Works offline on the laptop.

CAD to Cost

Measures the part, and asks you where the geometry cannot decide.

Self-audit

Checks every result: wrong machine size, wrong volume for tooling, odd cycle times.

Learning from actuals

Log a real quote or PO price. After 3 for a commodity, the band is corrected.

Supplier price monitor

Compares logged prices with the should-cost and flags where a supplier sits above it.

Optional AI Mode — Off at JLR

Needs AI approval. Turned on by a setting, with no rebuild.

Describe a part

Type a description and the AI suggests the inputs. The engine still does the sum.

PCB photo to BOM

Reads a board photo into a bill of materials, then costs it with the rules.

AI assistant

Answers costing questions in plain English.

AI agent

Works through a costing task step by step, when asked.

The common thread: AI never sets a price, on or off. Every number stays arithmetic you can check.

CHECKS AND PROOF

The Tool Checks Itself — and Says What Is Proven

It checks every result before you see it, shows how unsure it is, and learns from real prices.

It checks its own work

A self-audit runs on every result. It flags a wrong machine size, tooling spread over the wrong volume, a material-only costing, implausible cycle times, and more.

It shows how unsure it is

Every result carries a P10–P90 band from a Monte Carlo run over every cost driver, including flat material prices.

It learns from real prices

Log a real quote or PO price with "Log Actual £". After 3 actuals for a commodity, the band is corrected by real data.

What is proven — and what is not

Status as of September 2026.

Check	Evidence	What it shows	Status
Hand calculation	Reference bracket, £23.27	Engine arithmetic matches to <0.01%	Proven
Real-parts baseline	6 production parts pinned	A change that moves one fails the build	In place
Automated tests	2,438 tests + browser tests	Every commodity, exports, STL, accessibility	Every run
JLR-paid prices	None compared yet	Measured as actuals are logged	Not proven

Built to be checked, not just believed

- 2,438 automated tests — if the logic breaks, the build fails
- Browser tests cost every commodity and check accessibility on every run
- The CAD engine is tested inside the image we ship

REAL PARTS · REGRESSION BASELINE

Six Real Production Parts — Pinned in the Build

Measured from real CAD and costed by the product's own chain, at UK rates. Move one, and the build fails.

Part	What it is	Route (engineer's answer)	Should-cost	Measured	Why this route
Steering knuckle	Suspension upright, RH	Forging, steel	£ 32.16	9.0 mm bulk wall · 6% fill	Safety-critical, so forged; bore and faces machined after
Casting bracket	Mounting bracket	Cast + machine, steel	£ 37.61	10.2 mm wall · 16% fill · 10 holes	Named as a casting; holes and threads to finish
PRCR002 housing	Cast housing, 277 mm	Cast + machine, aluminium	£ 52.67	15.1 mm bulk wall · 9% fill	Assumed not pressure-tight — if it is, the cost rises
Part1	Machined part, 274 mm	Machining, aluminium	£ 38.93	10.4 mm bulk wall · 23% fill	Machined from billet at this volume
Seat locking bracket	Pressed bracket	Sheet metal, steel	£ 4.96	1.55 mm wall · 15 bends · 4% fill	The one genuine pressing
Spur gear m3 z38	38 teeth, ISO class 8	Gear cutting, steel	£ 25.50	Module 3, teeth counted off the tip	Routed by measuring the gear

UK region, GBP | Answers are stated engineering judgements, so they can be argued with | This pins what the tool says — none of these has been compared with a price JLR paid

REPORTING & DEPLOYMENT

Getting the Number Out — Safely

Reports people can act on, and a tool that is safe on a locked-down laptop.

Excel — Six Sheets

The whole build-up, not just the total.

Summary · material · operations ·
machine rates · labour rates ·
and a traceability sheet.

PDF Report and Negotiation Pack

The versions you send out.

Both carry the on-screen total to the penny.
Files are named <type>-<part>-<date>.

Windows Laptop Package

Copy a folder, double-click to start.

No installer, no admin rights, no internet.
Listens on the laptop only, unless shared.

Part Photos

Attach a photo to any calculation.

It appears on the cover of the PDF report,
so everyone knows which part it is.

Sign-In and Privacy

Sign-in on everything that holds data.

You see only your own saved scenarios.
A password reset or "sign out everywhere"
ends every old session.

Help and Accessibility

Nobody needs training to start.

Getting started, a glossary and
troubleshooting — all in the app.
Every page passes WCAG 2.1 AA (automated).

VALUE — HONESTLY STATED

What We Can Claim Today

Only what the repository can show. Accuracy against real prices is measured as they are logged.

<0.01% Engine vs hand calc, reference part	6 Real parts pinned in the build	2,438 Automated tests	19 Manufacturing processes	0 Estimates compared with a JLR-paid price, so far
Speed The engineer answers a few questions instead of building a spreadsheet. We will time it in the pilot.	Accuracy Arithmetic proven against a hand calc. Accuracy against real prices is measured as they are logged.	Negotiation A floor price you can defend, bucket by bucket. Supplier margin becomes visible instead of assumed.	Early Warning A cost from CAD at concept stage, while a change is still cheap.	
Where to Build It Compare 20 regions in one table. Make-vs-buy answered with real rates.	One Method The same model everywhere, with every assumption on record.	The Right Volume See what the part costs as volume grows, and when a new process starts to pay.	Safe to Run A Windows folder on a locked-down laptop. No internet, no admin rights, no AI.	

VISION & NEXT STEPS

What Comes Next

CostVision runs today. Here is what could come next — and a small first step.

Runs Today

- 19 manufacturing processes, plus assembly and software
- CAD to Cost on rules — asks where the geometry can't decide
- Rate library dated 16 June 2026, editable
- Self-audit and a P10-P90 band on every result
- Learns from logged quotes and PO prices
- 20 regions, each in its own currency
- Supplier quote comparison and negotiation pack
- Excel and PDF that match the screen to the penny
- Windows laptop package — no internet, no admin

Next

Connect to Teamcenter and SAP
Cost-target view for programme teams
Compare several suppliers side by side

Then

Calibrate from live ERP actuals
Draft the RFQ from the should-cost
Portfolio view for leadership

Later

Cost that updates as the design changes
Redesign suggestions from AI — only if AI is approved
Supplier risk scoring

Pilot — the Ask

- Two or three commodities, one programme team
- Four to six weeks, on real parts with real quotes and PO prices
- Log every actual, so accuracy is measured, not claimed
- Avinash Bhosale — Cost Engineering & Digital Innovation